

FIT-TRACK PRO

Universal Health & Fitness Management System

Installation, SSMS Setup & User Guide

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Document	FitTrack Pro — Complete Setup & User Manual
Version	1.0
Platform	Windows 10 / 11 (64-bit)
Database	Microsoft SQL Server Express (free)
Runtime	.NET 8 (free, from Microsoft)
Audience	Students and non-technical users

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1. What You Need

Before starting, check that your computer has the following.

Operating System	Windows 10 (version 1809 or later) or Windows 11
Processor	Any 64-bit processor — Intel or AMD
RAM	4 GB minimum, 8 GB recommended
Free Disk Space	3 GB (for SQL Server + app)
Internet	Required only during downloads
Account Type	Windows Administrator account

■ Important

You must right-click install.bat and choose Run as administrator. Running without administrator rights will fail at the build step.

2. Install .NET 8 Runtime

.NET 8 is the engine that runs FitTrack Pro. It is free and made by Microsoft.

1

Open your web browser.

Go to this address:

<https://dotnet.microsoft.com/download/dotnet/8.0>

2

Find the .NET 8.0 Runtime section.

Look for the row labelled ".NET Runtime 8.0.x".

Under Windows, click the button labelled x64.

3

Run the downloaded file.

It will be named: dotnet-runtime-8.0.x-win-x64.exe

Double-click it. Click Yes if Windows asks for permission.

4

Click Install and wait.

Installation takes about 1 minute.

When it says Installation was successful, click Close.

✓ You only need to install .NET 8 once. It stays on your computer permanently.

3. Install SQL Server Express

SQL Server Express is the free database engine that stores all your FitTrack data. It runs silently in the background as a Windows service.

- 1 Go to the Microsoft SQL Server download page:**
(Copy this address into your browser)

<https://www.microsoft.com/en-us/sql-server/sql-server-downloads>

- 2 Scroll down to find the Express edition.**
Click the Download now button under Express.
The file is named something like: SQL2022-SSEI-Expr.exe

- 3 Run the installer.**
Double-click the downloaded file.
Click Yes if Windows asks for permission.

- 4 Choose Basic installation type.**
On the first screen you see three choices.
Click Basic — this is the simplest option and all FitTrack needs.

- 5 Accept the licence and click Install.**
Installation takes 3 to 8 minutes. Do not close the window.
Your computer may slow down briefly — this is normal.

- 6 Note your instance name when installation finishes.**
The success screen shows your instance name.
It looks like: DESKTOP-ABC123\SQLEXPRESS
Write this down — you will need it in Section 6.

■ Important

The instance name is the address of your database server. Every computer has a different instance name. If you cannot find it: open Start Menu > search SQL Server Configuration Manager > click SQL Server Services > look at the name in brackets.

4. Install SQL Server Management Studio (SSMS)

SSMS is a free visual tool for managing your SQL Server database. You use it once to create the FitTrack database tables.

1 **Go to this address in your browser:**

The download starts automatically.

<https://aka.ms/ssmsfullsetup>

2 **Run the downloaded installer.**

File name: SSMS-Setup-ENU.exe

Double-click it. Click Yes for administrator permission.

3 **Click Install on the SSMS setup screen.**

Installation takes 5 to 15 minutes.

The progress bar may stop for a while — just wait.

4 **Click Restart if prompted, then installation is complete.**

SSMS is now ready to use.

✓ SSMS only needs to be installed once. After creating the database you rarely need to open it again.

5. Create the Database in SSMS

This section walks you through connecting SSMS to your SQL Server and running the script that creates all the FitTrack tables. You do this once.

5.1 Connect to SQL Server

- 1 Open SSMS.**
Click Start Menu > search SSMS or SQL Server Management Studio.
Click to open it.

- 2 The Connect to Server window appears automatically.**
If it does not appear: click File > Connect Object Explorer.

- 3 Fill in the connection details:**
Server type: Database Engine (already selected)
Server name: Your instance name from Section 3 Step 6
Example: DESKTOP-ABC123\SQLEXPRESS
Or try: localhost\SQLEXPRESS
Authentication: Windows Authentication (already selected)

- 4 Click Connect.**
If successful, the left panel (Object Explorer) shows your server name.
You are now connected to SQL Server.

■ Important

If Connect fails with an error: • Make sure SQL Server Express is running: Start Menu > search Services > find SQL Server (SQLEXPRESS) > right-click > Start. • Try typing just a dot (.) as the server name — it always means local machine.

5.2 Open and Run the Schema Script

The schema script creates the FitTrackOOP database and all the tables inside it.

- 1 In SSMS, click File > Open > File...**
A file browser opens.

- 2 Navigate to the FitTrackOOP folder you extracted from the zip.**
Open the folder: FitTrack > Database
Select the file: schema.sql
Click Open.

3

The script opens in the SSMS query editor.

You will see SQL code starting with: CREATE DATABASE FitTrackOOP

4

Click the Execute button (or press F5).

The button looks like a green play triangle in the toolbar.

Wait for the message at the bottom: Commands completed successfully.

■ **Important**

Do NOT run this script more than once. If you run it twice you will get an error saying the database already exists. That is fine — it means it was already created.

5.3 Verify the Database Was Created

1

In the Object Explorer panel on the left, click the arrow next to Databases.

You should see FitTrackOOP listed.

2

Expand FitTrackOOP > Tables.

You should see these tables:

Goals, Persons, ExerciseCategories, Exercises, WorkoutPlans, WorkoutPlanExercises, WorkoutSessions, SessionLogs, FoodCategories, FoodItems, NutritionLogs, ProgressSnapshots

✓ All 12 tables present means the database is ready. You can close SSMS now.

6. Configure the Connection String

The connection string tells the application where to find your SQL Server database. You need to set this before running the installer.

1 Open the FitTrackOOP folder you extracted from the zip file.

Find the FitTrack subfolder.

2 Open the file Program.cs in Notepad.

Right-click Program.cs > Open with > Notepad.

If Notepad is not listed, click Choose another app and find Notepad.

3 Find this line near the top of the file:

It is the first line with the word connectionString.

```
string connectionString = "Server=localhost\\SQLEXPRESS;Database=FitTrackOOP;Trusted_Connection=True;TrustServerCertificate=True;";
```

4

Replace localhost with your actual instance name if needed.

If your instance name from Section 3 was DESKTOP-ABC123\\SQLEXPRESS, change localhost to DESKTOP-ABC123

Leave everything else exactly as it is.

Example result:

```
string connectionString = "Server=DESKTOP-ABC123\\SQLEXPRESS;Database=FitTrackOOP;Trusted_Connection=True;TrustServerCertificate=True;";
```

5

Save the file.

Press Ctrl + S, then close Notepad.

✓ If you are unsure of your instance name, try leaving localhost — it works on most machines.

7. Run the Installer

The installer builds the application into a single .exe file and creates a desktop shortcut. Run it once after completing all previous sections.

1

Open the FitTrackOOP folder.

Find the file named install.bat

2

Right-click install.bat and choose Run as administrator.

Do NOT double-click — you must right-click and choose Run as administrator.
Click Yes if Windows asks for permission.

3

A black window opens and shows installation progress.

You will see numbered steps: [1/6] through [6/6]
Step 5 (building the application) takes 1 to 3 minutes.
Do not close the window.

4

Wait for the success message.

When installation is complete you will see:
Installation Complete!
Desktop shortcut: FitTrack Pro created
Press any key to close the window.

■ **Important**

If you see [ERROR] .NET Runtime is not installed — go back to Section 2. If you see [ERROR] SQL Server service not detected — go back to Section 3. If you see [ERROR] Build failed — check that you edited Program.cs correctly in Section 6.

8. Launch the Application

After installation, launching FitTrack Pro takes just a few seconds.

1

Find the FitTrack Pro shortcut on your desktop.

It was created automatically by the installer.

2

Double-click the shortcut.

A black terminal window opens.

The first launch loads and saves seed data (exercises, foods, goals).

This takes about 3 seconds.

3

You will see the main menu:

=== FIT-TRACK PRO ===

[1] Register [2] Login [0] Exit

The application is ready.

■ **Important**

The black window must stay open while you use the app. Closing it exits the application. Your data is saved to the database automatically — nothing is lost when you exit.

9. First Use — Register and Login

The first time you open the app you need to create an account.

Registering

1

Type 1 and press Enter to choose Register.

The app shows a list of fitness goals.

2

Note the goal number that matches your goal.

1 = Weight Loss 2 = Fat Loss 3 = Muscle Gain
4 = Weight Gain 5 = Maintain

3

Answer each question — type your answer and press Enter.

Full Name, Username (no spaces), Password, Email

Role: type Trainee (or Trainer or Admin)

Gender: type Male or Female or Other

Age, Height in cm (e.g. 175), Weight in kg (e.g. 72.5)

Body Fat %: press Enter to skip

Goal ID: type the number from step 2, or Enter to skip

✓ **Registration successful means you are now logged in automatically.**

Logging In (future sessions)

1

Type 2 and press Enter to choose Login.

Type your Username, press Enter.

2

Type your Password and press Enter.

You will see: OK — Welcome [Your Name]

10. Menu Guide

After logging in, type any number and press Enter to choose an option.

All Users — Trainee Menu	
[1] View Profile	Shows your name, role, goal, BMI, trainer, and account details.
[2] Update Profile	Change your name, email, weight, height, goal, and body fat.
[3] Change Password	Enter current password then new password to change it.
[4] Health Metrics	Choose your activity level. Shows BMI, BMR, TDEE, calorie target.
[5] All Goals	Lists all 5 goals: Weight Loss, Fat Loss, Muscle Gain, Weight Gain, Maintain.
[6] View Active Plan	Shows your current workout plan with exercises by day.
[7] List Exercises	Shows the full exercise library (40+ exercises with details).
[8] Log Workout	Record a workout: duration, exercises, sets, reps, weight used.
[9] Session History	Shows your last 10 completed workout sessions.
[10] Search Food	Search the food library by name (57 foods available).
[11] Log Meal	Record a meal: choose food, meal type, serving in grams.
[12] Daily Nutrition	Summary of today's total calories and macros. Enter a date or press Enter for today.
[13] Log Progress	Record your current weight and body fat. BMI is calculated automatically.
[14] Progress History	Shows your last 10 body measurement records over time.
[0] Logout	Return to the main login screen.

Trainer Extra Options	
[15] My Trainees	Lists all clients assigned to you by an Admin.
[16] Create Plan	Build a custom workout plan and optionally assign it to a client.
[17] Assign Trainer	Link a trainee account to a trainer account using their Person IDs.

Admin Extra Options	
[18] All Users	View every account in the system with IDs, roles, and goals.

[19] Set Role	Change any user's role to Trainee, Trainer, or Admin.
[20] All Plans	View every workout plan in the system.

11. Troubleshooting

Problem: SSMS says Cannot connect to server

Solution: Make sure SQL Server Express is running.

Open Start Menu > search Services > find SQL Server (SQLEXPRESS) > right-click > Start. Then try connecting in SSMS again.

Also try using a single dot (.) as the server name.

Problem: Installer says .NET Runtime is not installed

Solution: Go back to Section 2 and install .NET 8 Runtime. After installing, close all windows, open a new Command Prompt, and run install.bat again.

Problem: Installer says Build failed

Solution: The connection string in Program.cs may have a typo. Open Program.cs in Notepad and check the server name carefully. Also check install_log.txt in C:\Users\YourName\AppData\Local\FitTrackPro\ for the exact error.

Problem: App opens but crashes immediately with connection error

Solution: The database does not exist or the connection string is wrong. Re-check Section 5 (run schema.sql in SSMS) and Section 6 (connection string in Program.cs). Then run install.bat again to rebuild.

Problem: ERROR: Username already taken

Solution: Choose a different username. Each username must be unique in the system.

Problem: ERROR: User not found

Solution: The Person ID you typed does not exist. Use option [18] (Admin) to see all IDs.

Problem: ERROR: Food item not found / Exercise not found

Solution: Use option [10] to search foods or option [7] to list exercises. The correct ID is shown in square brackets next to each item.

Problem: Schema.sql gives error: Database already exists

Solution: This means you already ran it successfully. You do not need to run it again. Open SSMS, expand Databases, and verify FitTrackOOP is listed.

Problem: Forgot my password

Solution: An Admin account can delete your user (option [19] to find ID, then [0] logout and re-register). All your workout and nutrition data linked to that account will also be deleted.

Problem: Desktop shortcut does nothing / shows error

Solution: The app may have been moved or deleted. Run install.bat again as administrator to reinstall.

12. Uninstalling

To remove FitTrack Pro from your computer:

1

Open the original FitTrackOOP folder you extracted from the zip.

Find the file named uninstall.bat

2

Double-click uninstall.bat.

Type YES when prompted and press Enter.

3

The application files and desktop shortcut are removed.

Your database in SQL Server is NOT deleted automatically.

To also remove the database: open SSMS and run: DROP DATABASE FitTrackOOP;

✓ **Uninstall is complete. .NET and SQL Server remain on your PC — remove them via Windows Settings > Apps if needed.**